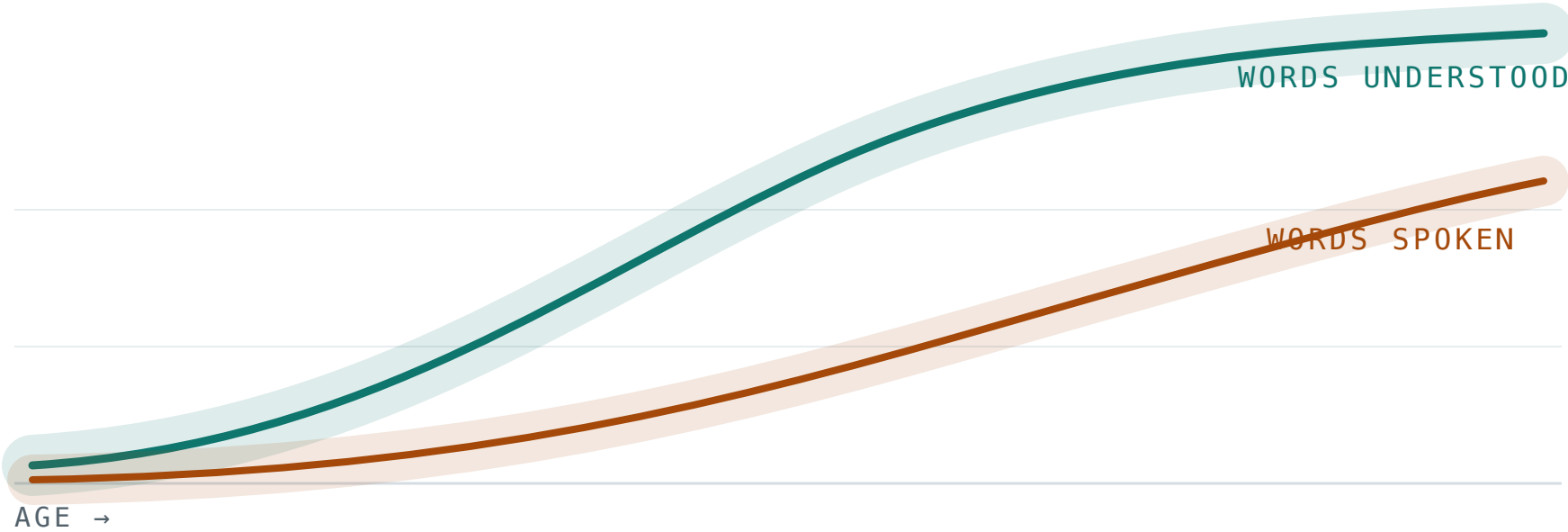


How children with Down syndrome build vocabulary

A pooled Bayesian study of the words children understand, say and sign, across eleven international studies.

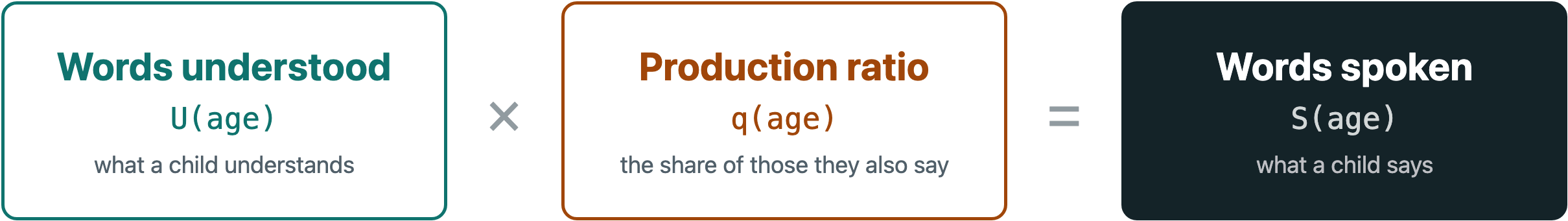
576 children · 11 international studies · ages 8 months to 9½ years · roughly 1,000 observations



SCHEMATIC · GROWTH CURVES WITH CREDIBLE BANDS · THE FITTED VERSIONS ARE INSIDE

How the vocabulary model works

The modelling idea at the centre of the study: spoken vocabulary is built from what a child understands and the share of it they say.



Modelling the two together means spoken vocabulary can **never exceed** what a child understands, and the comprehension-production gap falls out directly as the production ratio **q**.

THE DATA

Parent-report vocabulary checklists, **pooled across 11 international studies** onto a common 810-word scale: **576 children**, ages 8 months to about 9½ years, roughly 1,000 observations.

THE ESTIMATE



Each term is a **smooth curve over age** with full posterior uncertainty, so every result is a credible range, not a single number.

Hierarchical Beta-Binomial

Gaussian-process (HSGP) trends

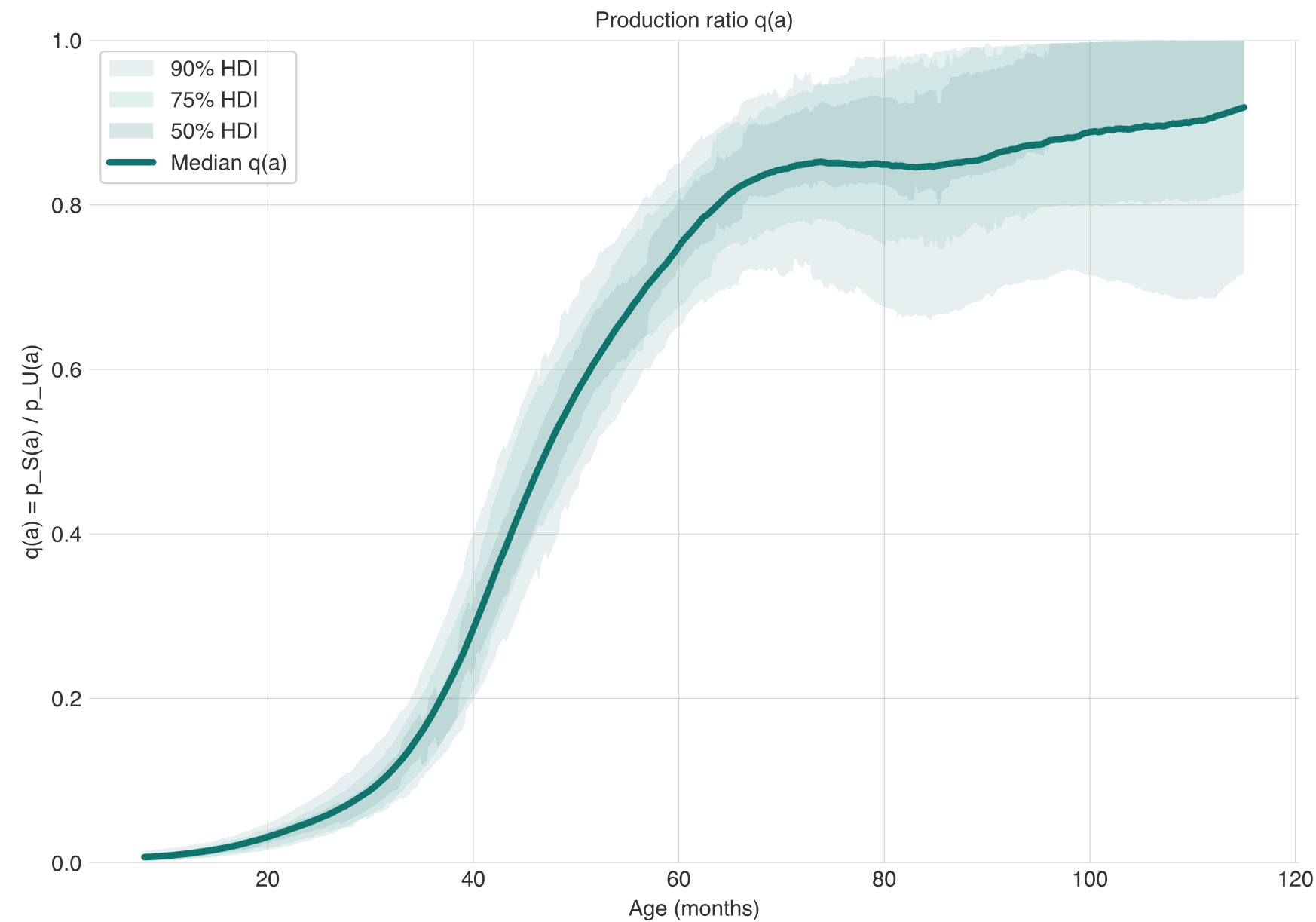
study & child random effects

PyMC / NUTS

PSIS-L00 comparison

Understanding runs years ahead of speaking

The share of understood words a child with Down syndrome also says, estimated across age from pooled international data.



Each child's **production ratio** is the fraction of the words they understand that they also say. Comprehension leads production for years, so the ratio stays low at first, then climbs steeply.

~1% **~52%** **~86%**
at 1 year at 4 years by 7½ years

Wider than in typically-developing children at low vocabulary levels, narrowing as vocabulary grows.

What the study finds

The four headline findings from the pooled fit.

- **Understanding runs years ahead of speaking**

The production ratio is the share of understood words a child also says. It climbs steeply with age, reaching roughly half by age four and about 86% by seven and a half.

- **Spoken vocabulary lags furthest**

By around age two, most children with Down syndrome say fewer words than the lowest-scoring tenth of typically-developing children. Understanding is delayed too, but by less.

- **The gap is wider than in typical development, then narrows**

At the same level of comprehension, children with Down syndrome say a smaller share of the words they know. The difference peaks at around 13 percentage points at 50–150 understood words, then fades as vocabulary grows.

- **Signing offers a modest early bridge**

Counting signed words credits children with expressive vocabulary they already have, and narrows the early gap somewhat. The narrowing is concentrated around ages two to three; the spoken-word gap stays large.

The findings are preliminary and observational; estimates come from development-tier model fits and will sharpen as more data arrive.